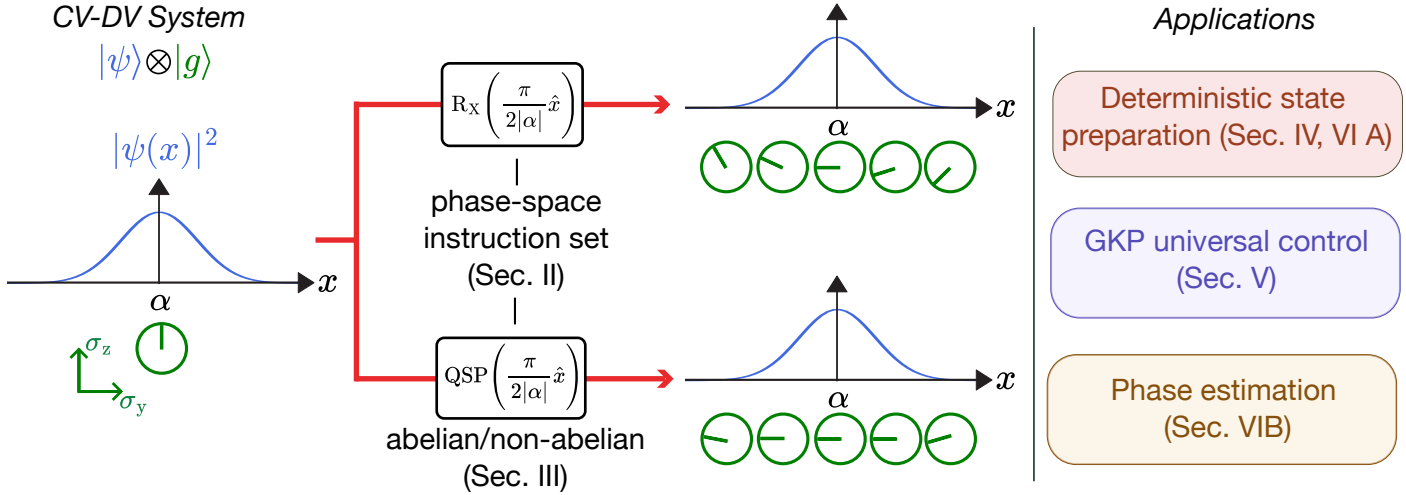
CV-DV System
$|\psi\rangle \otimes |g\rangle$
$|\psi(x)|^2$
x
α
σ_z
σ_y
$\mathrm{R_X}\left(\frac{\pi}{2|\alpha|}\hat{x}\right)$
phase-space instruction set (Sec. II)
$\mathrm{QSP}\left(\frac{\pi}{2|\alpha|}\hat{x}\right)$
abelian/non-abelian (Sec. III)
x
α
x
α
Applications
Deterministic state preparation (Sec. IV, VI A)
GKP universal control (Sec. V)
Phase estimation (Sec. VIB)